



Northstar Text-to-911 Middleware POC

Senior Forward Deployed Engineer Technical Presentation – RapidSOS

FASTAPI · PYTHON · HTML/CSS/JS · PYTEST · AI-ASSISTED DEV

The Challenge

Bridging Telecom & Emergency Dispatch

Build a lightweight middleware service connecting a fictional telecom provider – **Northstar** – to an operational text-to-911 agent console.

The system manages **inbound/outbound messaging**, session lifecycle state, retries, expiration handling, and operator workflows through a real-time dispatch interface.

Inbound Processing

Receive and deduplicate SMS messages from the carrier

Session Lifecycle

Maintain state, expiration, and continuity across caller sessions

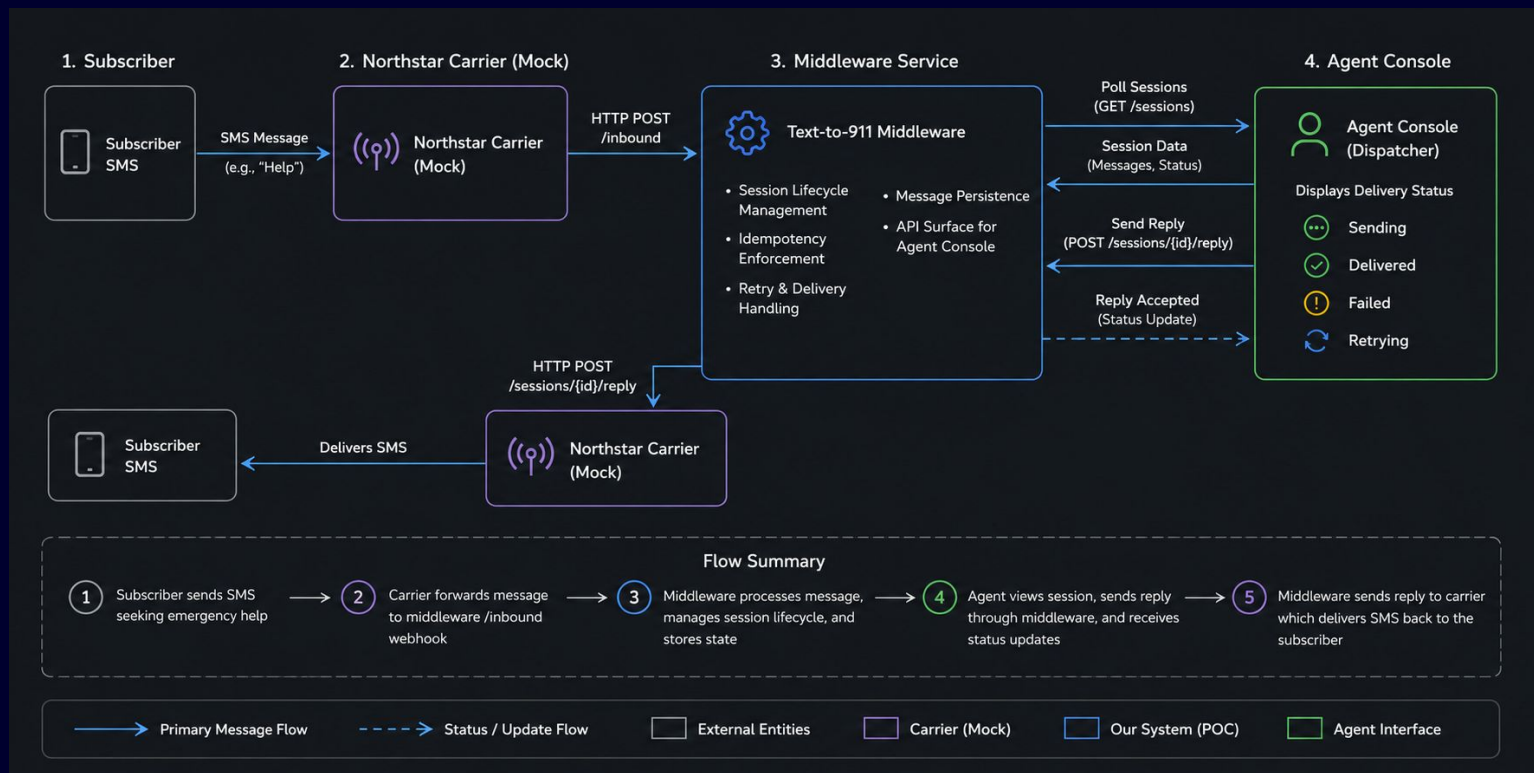
Agent Workflows

Support operator replies and provide a lightweight operational UI



Architecture Overview

System Structure



Middleware API Surface

Core operational endpoints powering session lifecycle and dispatcher workflows

 Session Configuration	GET	/config	Session TTL and expiring-soon thresholds (session_ttl_seconds, session_expiring_soon_seconds)
 Session Queue	GET	/sessions?group_by_phone=true	Grouped caller queue for operator sidebar (one row per caller / PhoneSummary)
	GET	/sessions?include_expired=true	Flat per-session list for debugging and tools (includes active and expired sessions)
 Session Detail	GET	/sessions/{id}	Thread messages, previous_sessions, is_reply_target, and delivery state fields
 Agency Operations	PATCH	/sessions/{id}/tags	Apply agency tags to the current session (FIRE, MEDICAL, POLICE)
 Outbound Messaging	POST	/sessions/{id}/reply	Send dispatcher reply Request: { "text", "timestamp"? } Response: { success, error, delivery_attempts, duplicate }
 Backend-enforced lifecycle state	 Idempotent workflows	 Delivery-state persistence	 Operational visibility for dispatchers



Session Lifecycle Design

State Management Philosophy



1



Session state follows SLA and is configurable

Session lifecycle follows configurable SLA timing through shared middleware policy controls.

2



Incorporated session state in various ways into both the messages and conversation panel

Session state is surfaced directly in the conversation panel through countdowns, banners, disabled reply states, and operator warnings.

3



Understanding agents work fast – we provide previous session states for subscriber context

Previous sessions remain visible inline so dispatchers retain historical caller context when new sessions are created.

4



We store and index session states, tags and conversation history in our data store

Session lifecycle state, tags, and conversation history are indexed for fast operational lookup and filtering workflows.



Operational Agent Console

Designed for Clarity Under Pressure

Agent Console

Text-to-911

STATUS

All

READ

All

AGENCY DEPLOYED

All

+15551230001

Sorry — are you still there?

May 25, 3:13 PM

+15551230009

PLEASE SEND HELP

May 25, 3:12 PM

+15551230011

911 Dispatch: POLICE are on the way. Stay aw...

May 25, 3:11 PM

+15551230010

911 Dispatch: Please share your location and ...

May 25, 3:10 PM

+15551230004

911 Dispatch: POLICE have been dispatched, ...

May 25, 3:10 PM

+15551230007

911 Dispatch: FIRE units are responding. Get ...

May 25, 3:09 PM

+15551230003

911 Dispatch: MEDICAL is responding. Get ...

May 25, 3:09 PM

+15551230001

ACTIVE Expires 4:55 remaining

Agent - May 25, 3:06 PM

911 Dispatch: Message Received. Please describe what is happening so we can dispatch the appropriate emergency services.

User - May 25, 3:06 PM

My house is on fire!

Agent - May 25, 3:06 PM

FIRE units are on the way. Stay outside and keep clear of the structure.

Agent - May 25, 3:06 PM

911 Dispatch: FIRE units should be on scene. Is everyone out safely, and is the situation under control?

User - May 25, 3:07 PM

The fire trucks just got here. We're all outside the house.

Agent - May 25, 3:07 PM

911 Dispatch: Good. Stay clear of the structure and let us know if anything changes.

User - May 25, 3:07 PM

Yes — thank you so much for your help.

Agent - May 25, 3:07 PM

911 Dispatch: You're welcome. We're glad you're safe. Text us again anytime you need emergency assistance.

Current session — replies go to this session

User - May 25, 3:13 PM

Sorry — are you still there?

AGENCY DEPLOYED

☐ Fire

☐ Medical

☐ Police

Type agent reply...

Send



1. Session Filtering

Filter sessions by status, read status, and agency deployment to quickly focus on what matters.



2. Number Grouping

Sessions are grouped by phone number. Each item shows session state, text preview, last action date, and unread indicator.

● Active ● Expiring/Soon ● Expired ● Unread



3. Session State & Countdown

Current session state is always visible with a countdown timer to help operators act before expiration.



4. History Preservation & Separation

Expired sessions are preserved and separated from active conversations to maintain historical context without causing operational confusion.



5. Agency Deployed Session Tagging

Tag sessions by deployed agencies (Fire, Medical, Police) to streamline responsibility and coordination

☐ Fire ☐ Medical ☐ Police



Key Design Decisions & POC Enhancements

Beyond the Brief — Operation Design & Product Thinking



Backend as Source of Truth

Core business logic is enforced within the middleware service, including session lifecycle management, idempotency enforcement, retry orchestration, and delivery state handling.



Session Continuity

Sessions are grouped by caller phone number, and historical conversations remain preserved inline, allowing operators to maintain context across rollover and expired session workflows.



Operational Visibility

Lifecycle state, delivery feedback, unread indicators, countdown timers, and operational banners provide fast comprehension for dispatchers working under pressure.



Lightweight Reliability

The middleware abstracts retry handling, duplicate protection, and error recovery workflows behind the scenes so operators only see actionable operational state.



Future Improvements

Production Considerations



Product & Operational Enhancements

- Device geolocation metadata for dispatch awareness and caller routing
- AI-assisted summaries for repeat-caller continuity and cognitive load reduction
- Reporting and analytics support for operational metrics and workflow visibility



Prompt Exclusions / Production Hardening

- Persistent datastore and indexed idempotency keys
- Add authentication on API services
- Real-time push using SSE/WebSockets
- Structured observability and PSAP dashboards
- High availability deployment with stateless middleware replicas



Live Demo & Operational Walkthrough

End-to-end session lifecycle workflows and dispatcher interactions

- Live execution of 11 emergency response scenarios through the operational harness
- Simulated subscriber and dispatcher conversations running in parallel
- Inbound messages staggered to simulate concurrent operational workflows
- Real-time UI updates: unread indicators, lifecycle states, timers, and delivery feedback
- Sessions expire after TTL windows; new sessions are created when subscribers message after expiration

Scenario Coverage

#	Scenario	What It Demonstrates
1	Happy Path	Primary emergency dispatch workflow, resolution, and new session after TTL
2	Expired Session	Backend-enforced expiration and blocked reply behavior after TTL
3	Duplicate Outbound	Outbound idempotency handling during topology changes
4	Police Break-In	Urgent police dispatch workflow
5	Multi-Agency	Police and medical coordination within the same session
6	Medical Chest Pain	Medical dispatch workflow
7	Fire Smoke Alarm	Fire workflow with post-expiration idle
8	Outbound Failure	Retry orchestration and delivery recovery using FORCE FAIL
9	Fire and Medical	Combined fire and medical dispatch
10	Expiring Scenario	Operator visibility during expiring session workflows
11	Session Rollover	Session rollover and continuity handling



Q/A & DEMO

